

No. of Functional Features Included in the Solution

Date	15 July 2026
Team ID	SWTID-2026-7750
Project Name	Credit Card Approval Prediction System
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	Data Collection	Loads applicant and credit history datasets for model training.	Data Collection	Done	High
2	Data Preprocessing	Cleans the dataset, handles missing values, and removes unnecessary data.	prepare_data.py	Done	High
3	Feature Engineering	Creates AGE and YEARS_EMPLOYED features for better prediction accuracy.	prepare_data.py	Done	High
4	Categorical Data Encoding	Encodes categorical features using LabelEncoder.	train_model.py	Done	Medium
5	Class Balancing	Applies SMOTE to balance the dataset before model training.	train_model.py	Done	Medium
6	Model Training	Trains Logistic Regression, Decision Tree, Random Forest, and XGBoost models.	train_model.py	Done	High
7	Model Evaluation	Compares models using Accuracy, Precision, Recall, F1-Score, Confusion Matrix, and ROC-AUC.	train_model.py	Done	High
8	Web-Based Prediction System	Allows users to enter applicant details and receive approval predictions with a confidence score through a Flask web application.	app.py	Done	High

Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	8
Additional / Nice-to-Have Features	0
Features Tested & Verified	8

Feature Category Breakdown:

S No	Category	Features in Category	Example Features
1	User Interface (UI)	2	Applicant Details Form, Prediction Result Page
2	Backend / Logic	4	Data Preprocessing, Feature Engineering, Model Training, Prediction
3	Database / Storage	1	CSV Datasets, Trained Model (model.pkl), Encoders
4	API / Integration	1	Flask Application Integration
5	Security / Authentication	0	Not Implemented